# Ultraviolet Catastrophe, Planck's Quantization and Einstein's Light Quanta:

## What Classical Physics Could Not Explain — and How T0 Theory Resolves Both Aspects Geometrically

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# Abstract

The ultraviolet catastrophe and Einstein's light quantum hypothesis mark two of the most profound crises and breakthroughs in the history of physics. This document analyses both aspects from three perspectives: first, the historical and physical development — from Maxwell's electromagnetic wave theory through Planck's desperate quantization trick to Einstein's thermodynamically forced conclusion that light consists of independent energy quanta; second, the remaining open questions in standard physics — why $\epsilon = hf$? why does $h$ have this particular value? what sets an upper limit on frequency?; and third, the T0 answers to both questions: Planck's constant $h$ emerges without free parameters from the geometric parameter $\xi = 4/30000$, and the ultraviolet divergence is not a mathematical artefact but a symptom of the false assumption that spacetime is continuous down to arbitrarily small scales. The sub-Planck scale $L_0 = \xi \cdot \ell_P$ imposes a geometrically enforced maximum frequency $f_{\text{max}}$ above which no modes exist.

# .1 Introduction: Two Crises, One Geometric Principle

## The Historical Context

At the end of the 19th century, physics found itself in a peculiar combination of triumph and bewilderment. Maxwell's theory of electromagnetism had unified light, electricity, and magnetism in a single, mathematically elegant framework. Phenomena such as interference and diffraction seemed to confirm the wave picture of light beyond doubt. Many physicists regarded Maxwell's theory as one of the pinnacles of scientific achievement.

Yet this very theory failed spectacularly the moment it was applied to one of the simplest imaginable situations: the thermal radiation of hot bodies. The result was a prediction that was physically absurd — infinite energy density.

This crisis became the birthplace of quantum physics.

## What This Document Achieves

We follow three parallel threads throughout this document:

1. **The classical crisis**: Why does classical theory diverge at high frequencies? What exactly goes wrong, and why is this not trivial?
2. **The historical answers**: How did Planck (1900) and Einstein (1905) solve the problem — and what remained open? It is shown that Einstein did *not postulate* the light quantum but *extracted* it from thermodynamics.
3. **The T0 perspective**: How does T0 theory explain both aspects geometrically? Planck's constant emerges from $\xi$, and the UV divergence disappears because spacetime itself possesses a geometric minimum structure.

**Guiding Questions**

Why does $\epsilon = hf$ hold? Why does Planck's constant have exactly this value? And why is there a natural upper limit to frequency? Classical physics cannot answer these questions. T0 can.

## .2 The Ultraviolet Catastrophe: A Precise Diagnosis

### The Blackbody Radiation Problem

Imagine a completely closed cavity whose walls have been heated uniformly to temperature $T$ and are in thermal equilibrium. If a tiny hole is drilled in the wall, radiation escapes. This *cavity radiation* (also: *blackbody radiation*) possesses a remarkable property: its spectrum depends *solely* on the temperature — not on the material or shape of the cavity.

The experimentally measured quantity is the *spectral energy density* $\rho(f,T)$: the energy per unit volume per unit frequency interval. At different temperatures, one obtains characteristic curves with clear signatures:

- The peak of the curve shifts to higher frequencies as temperature increases (Wien's displacement law).
- The total area under the curve grows as the fourth power of temperature (Stefan-Boltzmann law: $u \propto T^4$).
- A hot piece of metal glows first dull red, then brighter orange-yellow, finally white-hot — the dominant frequency visibly shifts upward.

The challenge for theoretical physics around 1900 was precise: derive this curve from the fundamental laws of physics.

### The Classical Approach and Its Failure

The natural starting point was Maxwell's theory combined with the well-established statistical mechanics. The approach is conceptually simple:

**Step 1: Counting modes.** Inside the cavity, only standing electromagnetic waves can form — precisely those wavelengths for which an integer number of half-wavelengths fit between the walls. The number of electromagnetic modes per unit volume in the frequency interval $[f, f+df]$ is:

$$g(f)\,df = \frac{8\pi f^2}{c^3}\,df \tag{1}$$

The crucial point: mode density grows quadratically with frequency. For every doubling of frequency, there are four times as many independent wave patterns. And there is *no upper limit* — wavelength can become arbitrarily small, frequency arbitrarily large.

**Step 2: Equipartition theorem.** Classical statistical mechanics (Boltzmann, Maxwell) yields the equipartition theorem: in thermal equilibrium at temperature $T$, each quadratic degree of freedom carries on average the energy $\frac{1}{2}k_BT$. An electromagnetic wave has two quadratic degrees of freedom (electric and magnetic field), so each mode carries on average:

$$\langle E_{\text{mode}} \rangle = k_BT \tag{2}$$

This holds *independently of the frequency of the mode*. A mode at $10^{15}$ Hz receives the same average energy as a mode at $10^3$ Hz.

**Step 3: Spectral energy density.** The spectral energy density follows as a simple product:

$$\rho_{\text{class}}(f,T) = g(f) \cdot \langle E_{\text{mode}} \rangle = \frac{8\pi f^2}{c^3} \cdot k_BT \tag{3}$$

This is the *Rayleigh-Jeans law*. It agrees well with experiment at low frequencies. At high frequencies, however, it diverges catastrophically.

## The Scale of the Catastrophe

The total energy density is obtained by integrating over all frequencies:

$$u_{\text{class}} = \int_0^\infty \rho_{\text{class}}(f,T)\,df = \frac{8\pi k_BT}{c^3}\int_0^\infty f^2\,df = \infty \tag{4}$$

The integral diverges. Classical physics predicts: a cavity in thermal equilibrium contains *infinite energy*. This is physically absurd.

This divergence is called the **ultraviolet catastrophe** because it manifests most dramatically at the high-frequency (ultraviolet and beyond) end of the spectrum. The name is somewhat misleading —

the divergence actually extends across the entire high-frequency spectrum, not just the ultraviolet range.

**Why the Failure Is Deep**

The Rayleigh-Jeans law is not a calculational error. It is the correct application of two well-confirmed theories: Maxwell's electrodynamics and Boltzmann's statistical mechanics. Its failure therefore shows that one of these foundations must be fundamentally wrong — or that an important physical fact is missing.

## The Two Empirical Partial Solutions

Before the complete solution, there were two important empirical clues:

**The Rayleigh-Jeans law** (derived around 1900) fits well at *low* frequencies:

$$\rho_{\mathrm{RJ}}(f,T) \approx \frac{8\pi f^2}{c^3} k_B T \quad \text{(for } f \to 0\text{)} \tag{5}$$

**Wien's radiation law** (1896) describes *high* frequencies well:

$$\rho_{\mathrm{Wien}}(f,T) = a f^3 \exp\left(-\frac{bf}{T}\right) \tag{6}$$

where $a$ and $b$ are empirical constants. At high frequencies, the exponential falls off faster than $f^2$ grows — the integral converges.

The problem: both laws hold only in their respective limiting regimes. Between them lie the experimentally determined data, showing a smooth connection. A unified theory was missing.

## .3 Planck's Solution: An Act of Desperation

### Planck's Strategy

Max Planck approached the problem differently from his contemporaries. Rather than starting from a specific microscopic model, he worked *thermodynamically backwards*: he searched for the mathematical expression for the entropy of cavity radiation that was consistent with the empirically known spectrum.

The decisive step: Planck found a formula that reduces to the Rayleigh-Jeans law at low frequencies and to Wien's law at high frequencies:

$$\rho_{\text{Planck}}(f, T) = \frac{8\pi h f^3}{c^3} \cdot \frac{1}{\exp\left(\frac{hf}{k_B T}\right) - 1} \tag{7}$$

This formula describes the experimental spectrum across the entire frequency range with extraordinary precision — with only one new constant: Planck's constant $h$.

### The Price: Quantization as a Postulate

To derive his formula, Planck had to make an assumption that he himself described as an *act of desperation*: he assumed that energy can only be exchanged in *discrete multiples* of $hf$. The energy of a single portion (*quantum*) is:

$$\epsilon = hf \tag{8}$$

With this assumption, the average energy of a mode at frequency $f$ is no longer $k_B T$ (equipartition theorem) but:

$$\langle E_{\text{mode}} \rangle = \frac{hf}{\exp\left(\frac{hf}{k_B T}\right) - 1} \tag{9}$$

For $hf \ll k_B T$ (low frequencies), this expression approaches $k_B T$ — the classical result is recovered. For $hf \gg k_B T$ (high frequencies), the average energy falls off exponentially — exactly what is needed to suppress the divergence.

## What Planck Did Not Explain

Quantization solved the spectral problem mathematically. But Planck himself regarded it as a computational auxiliary construct, not a physical statement about the nature of light. In his view, the electromagnetic field remained a continuous wave. Quantization was a property of the *interaction* between matter and radiation — not of radiation itself.

**Planck's Open Question**

Planck had found the curve. But he had not explained *why* energy is exchanged in discrete portions. The number $h$ was a free parameter — measured from the data, not derived from deeper principles. What is $h$? Why exactly this value?

## .4 Einstein's Light Quanta: Thermodynamically Forced

### A Widespread Misconception

Einstein's 1905 contribution is typically summarized as "the explanation of the photoelectric effect." This is misleading. The photoelectric effect appears only at the end of the paper as one of several consequences. Einstein does not begin with metals, electrons, or experiments.

He begins with a *structural problem within physics itself*.

### The Structural Problem

On one side stands the theory of matter: gases and other ponderable bodies are described by a *finite* number of discrete quantities — the positions and velocities of atoms. The state of the system is specified by a finite list.

On the other side stands Maxwell's theory: the state of the electromagnetic field is *not* given by a finite list but by continuous fields — smooth functions defined at *every* point in space. These are infinitely many degrees of freedom.

In Einstein's own words: there is a *profound formal distinction* between these two ways of describing nature. The question is: if both systems can be in thermal equilibrium — do they really have such a fundamentally different structure?

### Einstein's Method: Thermodynamics Instead of Model

Einstein refused to start from a speculative microscopic model of light. Instead, he wanted to discover what the most reliable available facts tell us about the nature of light.

These facts came from two sources:

1. The experimentally confirmed blackbody spectrum (particularly Wien's law in the high-frequency regime).
2. The most general and powerful equilibrium principle available: the second law of thermodynamics — specifically Boltzmann's statistical interpretation.

## The Thermodynamic Derivation

**Step 1: Entropy of radiation.** Einstein introduces a *spectral entropy density* $\sigma(f,T)$ — analogous to the spectral energy density $\rho(f,T)$. In thermal equilibrium, the thermodynamic relation holds:

$$\frac{\partial\sigma}{\partial\rho} = \frac{1}{T} \tag{10}$$

This is not a model assumption but a fundamental thermodynamic identity.

**Step 2: Entropy from Wien's law.** By substituting Wien's law (6) into the thermodynamic relation (10) and integrating, Einstein obtains the entropy density in the Wien regime:

$$\sigma(f,T) = -\frac{\rho}{bf}\left[\ln\left(\frac{\rho}{af^3}\right) - 1\right] \tag{11}$$

**Step 3: Entropy change under volume change.** Einstein now considers a narrow, quasi-monochromatic slice of radiation with total energy $E$ at frequency $f$, enclosed in volume $V_0$. He calculates the entropy change when this volume is changed to $V$:

$$\Delta S = S(V) - S(V_0) = \frac{E}{bf}\ln\left(\frac{V}{V_0}\right) \tag{12}$$

**Step 4: Comparison with an ideal gas.** This is the decisive moment. For an ideal gas of $N$ independently moving particles, Boltzmann's entropy formula $S = k_B \ln W$ (with $W$ as the number of microstates) yields:

$$\Delta S_{\text{gas}} = Nk_B \ln\left(\frac{V}{V_0}\right) \tag{13}$$

The two expressions (12) and (13) have *identical structure*. Comparing the exponents gives:

$$\frac{E}{bf} = Nk_B \quad \Longrightarrow \quad N = \frac{E}{k_B bf} \tag{14}$$

**Step 5: Energy of one quantum.** If the total energy $E$ is carried by $N$ independent entities, each carrying energy $\epsilon$, then $E = N\epsilon$, so:

$$\epsilon = \frac{E}{N} = k_B b f \tag{15}$$

Since Wien's constant $b$ turns out to be $b = h/k_B$ by comparison with Planck's law, we finally obtain:

$$\boxed{\epsilon = hf} \tag{16}$$

**Einstein's Decisive Insight**

The light quantum was not postulated. It was *forced* by thermodynamics. The only inputs were: (1) the empirical Wien law, (2) the thermodynamic definition of equilibrium, (3) Boltzmann's entropy-probability relation. Nowhere was a particle nature of light assumed — it emerged as a logical consequence.

## The Photoelectric Effect as Confirmation

Only after this derivation does Einstein apply his hypothesis to the photoelectric effect. The result is the famous equation:

$$E_{\text{kin, max}} = hf - \Phi \tag{17}$$

where $\Phi$ is the *work function* of the metal. This equation is linear in $f$ with slope $h$ — a very sharp, experimentally testable prediction.

Experimental confirmation came through Robert Millikan (1916), who despite years of confirming the equation doubted the light quantum picture. Planck, recommending Einstein for the Prussian Academy in 1913, praised his brilliance while cautioning that the light quantum hypothesis should be counted as a miss. The Nobel Prize (1921) honoured the equation — not the interpretation behind it.

## What Einstein Left Open

Even after Einstein's work, fundamental questions remained unanswered:

- **Why $\epsilon = hf$?** Einstein showed that monochromatic radiation in the Wien regime behaves thermodynamically as if it consisted of independent quanta. But *what* are these quanta microscopically?
- **Why exactly this value of $h$?** Planck's constant is $h \approx 6.626 \times 10^{-34}$ J·s. Why this numerical value? Where does it come from? In standard physics, $h$ is a free parameter — measured, not derived.
- **What limits frequency from above?** The UV divergence is suppressed by Planck's quantization, but the fundamental question remains: is frequency space really open all the way to $f \to \infty$? Is there no physical limit?

## .5 T0 Perspective I: The Origin of $\epsilon = hf$

### The Geometric Parameter $\xi$

T0 theory postulates that all physical constants emerge from a single dimensionless geometric parameter:

$$\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4} \tag{18}$$

This parameter describes the ratio between the fundamental sub-Planck scale and the cosmic scale. It is not free — it follows from the geometry of three-dimensional space at the Planck scale.

### The Binding Condition

The heart of T0 theory is time-mass duality:

$$T(x,t) \cdot m(x,t) = 1 \quad \text{(in natural units)} \tag{19}$$

The intrinsic time field of a particle with mass $m$ is:

$$T(x,t) = \frac{\hbar}{m(x,t) \cdot c^2} \tag{20}$$

Time and mass are not independent quantities — they are reciprocal aspects of a single geometric reality.

### What Happens to the Photon?

The photon has no rest mass: $m_\gamma = 0$. In the T0 binding condition $T \cdot m = 1$, this naively implies $T \to \infty$. The physical interpretation: the photon *is* its time field — it does not exist *in* time, it *is* a time-like structure. From the photon's perspective, no proper time elapses.

For a photon with frequency $f$:

$$E_\gamma = hf = \frac{\hbar}{T(x,t)_\gamma} \tag{21}$$

In T0, this is not an external equation but a consequence of the binding condition: energy is the reciprocal intrinsic time of the field.

### Planck's Constant as an Emergent Quantity

In standard physics, $h$ is a fundamental constant with value $6.626 \times 10^{-34}$ J·s. In T0, $h$ is not a free constant but emerges from $\xi$:

$$h = \xi \cdot \hbar_{\text{Planck}} \cdot f(\text{geometry}) \tag{22}$$

Concretely: the 2019 SI reform fixed $h$ by definition (when calibrating $\alpha = 1$ in rational T0 units). The numerical value of $h$ is not arbitrary — it encodes the geometric relationship between the sub-Planck scale $L_0 = \xi \cdot \ell_P$ and the macroscopic world.

**T0 Answer to Question 1**

Why does $\epsilon = hf$ hold? In T0, the answer is: because the energy of a photon equals the reciprocal intrinsic time of its field ($E = 1/T$), and frequency is the inverse time ($f = 1/T$). The equation $\epsilon = hf$ is not a postulate — it is the direct consequence of the time-field binding condition $T \cdot E = 1$ for massless fields.

### The Connection to Einstein's Argument

Einstein's thermodynamic derivation and T0's geometric derivation converge on the same result through two different paths:

**Table 1:** Two paths to the equation $\epsilon = hf$

| Aspect | Einstein 1905 | T0 Theory |
|---|---|---|
| Starting point | Entropy of Wien radiation | Time-field binding $T \cdot E = 1$ |
| Method | Thermodynamic comparison | Geometric deduction |
| Result | $\epsilon = hf$ (thermodynamically forced) | $\epsilon = hf$ (geometrically emergent) |
| Status of $h$ | Free fit parameter | Emerges from $\xi = 4/30000$ |
| What remains open | Why this value of $h$? | Everything explained |

## .6 T0 Perspective II: Why the UV Divergence Disappears

### The Classical Dilemma Revisited

The ultraviolet catastrophe arises from the combination of two facts:

1. The number of modes grows like $f^2$ — without upper bound.
2. Each mode receives the same average energy $k_B T$.

Planck's solution suppresses the divergence by making the average energy fall off exponentially at high frequencies. But the more fundamental question remains unanswered: *Is frequency space really physically meaningful all the way to $f \to \infty$?*

The implicit assumption of classical physics — and also of quantum field theory — is: yes. Frequency space is unbounded. Spacetime is continuous down to arbitrarily small scales.

T0 says: this assumption is wrong.

### The Sub-Planck Scale in T0

The geometric parameter $\xi$ defines a fundamental minimum structure of spacetime below the Planck scale:

$$L_0 = \xi \cdot \ell_P = \frac{4}{30000} \cdot \ell_P \approx 1.333 \times 10^{-4} \cdot \ell_P \tag{23}$$

$$T_0 = \xi \cdot t_P = \frac{4}{30000} \cdot t_P \approx 1.333 \times 10^{-4} \cdot t_P \tag{24}$$

where $\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m and $t_P = \ell_P/c \approx 5.39 \times 10^{-44}$ s are the ordinary Planck units.

The scale $L_0$ is the geometrically smallest length at which spacetime still carries well-defined degrees of freedom. Below $L_0$, no structure exists — not because we cannot measure it, but because spacetime itself carries no further degrees of freedom there.

### The Maximum Physical Frequency

If $T_0$ is the smallest physically meaningful time scale, then there is a maximum frequency:

$$f_{\max} = \frac{1}{T_0} = \frac{c}{L_0} = \frac{c}{\xi \cdot \ell_P} \tag{25}$$

Numerically:

$$f_{\max} = \frac{c}{\xi \cdot \ell_P} = \frac{3 \times 10^8 \text{ m/s}}{1.333 \times 10^{-4} \cdot 1.616 \times 10^{-35} \text{ m}} \approx 1.39 \times 10^{47} \text{ Hz} \tag{26}$$

This is $1/\xi$ times larger than the Planck frequency $f_P \approx 1.85 \times 10^{43}$ Hz, and is therefore finite. The decisive point: there exists a finite upper bound.

## The Integral Converges

In T0, the frequency integral is no longer improper:

$$u_{\text{T0}} = \int_0^{f_{\max}} \rho(f, T)\, df < \infty \tag{27}$$

The divergence from (4) disappears not through a mathematical trick but because the upper limit is physically real. Above $f_{\max}$, there simply are no modes — spacetime carries no electromagnetic degrees of freedom there.

**The Difference from Planck's Solution**

Planck suppresses the divergence at high frequencies through a decaying Boltzmann factor: $\langle E \rangle \propto \exp(-hf/k_B T) \to 0$. The integral converges because the integrand becomes exponentially small.

T0 eliminates the divergence in a fundamentally different way: there are no modes above $f_{\max}$. The upper integration limit is finite, not infinite. The divergence is not a mathematical artefact to be suppressed — it physically does not exist.

## Why Classical Theory Gets It Wrong

The classical Rayleigh-Jeans integral diverges because it starts from a false physical premise: spacetime is a continuous medium carrying infinitely many degrees of freedom at arbitrarily small scales.

T0 identifies this assumption as the true cause of the catastrophe:

$$\text{UV catastrophe} = \text{false assumption: } L_{\text{min}} = 0 \tag{28}$$

The correct statement is:

$$L_{\text{min}} = L_0 = \xi \cdot \ell_P \neq 0 \tag{29}$$

Spacetime is *discrete* at scale $L_0$ — not in the sense of a lattice model, but in the sense of a fractal geometric structure that contains no further degrees of freedom below $L_0$.

## Fractal Spacetime and Natural Regularization

T0 spacetime is fractal with Hausdorff dimension:

$$D_{\text{frak}} = 3 - \xi \approx 2.999867 \tag{30}$$

This slight deviation from 3 has profound consequences. In a fractal spacetime, the number of modes does not grow exactly like $f^2$ but like:

$$g_{\text{T0}}(f)\,df = \frac{8\pi f^2}{c^3} \cdot \left(\frac{f}{f_{\text{max}}}\right)^{-\xi \cdot D_{\text{frak}}} df \tag{31}$$

At frequencies far below $f_{\text{max}}$, this correction factor is $\approx 1$ — the classical mode density is reproduced. Near $f_{\text{max}}$, the factor provides additional suppression that coincides with the geometric boundary.

## .7 Synthesis: What T0 Achieves and What Remains Open

### The Complete Table

**Table 2:** Comparison: Classical Physics — Planck/Einstein — T0

| Problem | Classical | Planck 1900 | Einstein 1905 | T0 |
|---|---|---|---|---|
| UV divergence | Diverges | Suppressed (quantization) | Not addressed | Does not exist ($f_{\text{max}}$) |
| Spectrum | Wrong law | Correct law | Confirmed | $f_{\text{max}}$ correction |
| Status of $h$ | No $h$ | Free parameter | Free parameter | Emerges from $\xi$ |
| Why $\epsilon = hf$? | No answer | Postulate | Thermodynamically forced | Geometrically ($T \cdot E = 1$) |
| Maximum frequency | None | None | None | $f_{\text{max}} = c/(\xi \cdot \ell_P)$ |
| Nature of photon | Wave | Quantized | Quantized | Time-field state |
| Free parameters | Many | $h$ free | $h$ free | Zero |

### Why T0 Goes Deeper

Planck and Einstein treated the *symptom*. Planck through postulating quantization, Einstein through thermodynamic proof of quantum structure. But neither could explain:

- *Why* energy is divided into quanta $hf$.
- *Why* $h$ has exactly this numerical value.
- *What* is the physical limit of frequency space.

T0 addresses all three questions through a single mechanism: the geometric structure of spacetime at the sub-Planck scale $L_0 = \xi \cdot \ell_P$.

### Testable Predictions

The T0 answers are not speculative — they yield concrete, experimentally testable deviations:

1. **Planck spectrum with $f_{\text{max}}$ cutoff**: At extremely high energies, the blackbody spectrum should show a hard cutoff at $f_{\text{max}} \approx 10^{47}$ Hz. This lies far beyond current experimental capabilities, but is in principle testable.
2. **Fractal corrections in g-2**: The correction factor $K_{\text{frak}}^{3/2}$ of the T0 time field appears in the anomalous magnetic dipole moment calculation and reduces the theoretical deviation from experiment to $0.014\%$.

3. **Photoelectric effect correction**: At very high photon frequencies, Einstein's equation (17) should be modified by a $\xi$ correction factor:

$$E_{\text{kin, max}}^{\text{T0}} = hf\left(1 - \xi \cdot \frac{f}{f_{\text{max}}}\right) - \Phi \tag{32}$$

## The Methodological Parallel

It is remarkable that Einstein and T0 proceed methodologically identically. Einstein refused to presuppose a microscopic model of light. He followed thermodynamics wherever it led. The light quantum was the result, not the assumption.

T0 proceeds the same way. No specific model of spacetime discreteness is postulated. The question is asked: what follows from the geometric binding condition $T \cdot m = 1$ and the parameter $\xi$? The maximum frequency, the emergent Planck constant, and the suppression of the UV divergence are the results — not the assumptions.

**The Unified View**

The ultraviolet catastrophe and the Planck-Einstein quantum problem are *the same problem at two different levels*. At the energy level: why does $\epsilon = hf$ hold? At the frequency level: why is there an upper limit? In T0, both have the same answer: because spacetime at the scale $L_0 = \xi \cdot \ell_P$ possesses a geometrically defined fundamental structure that permits neither $h$ as a free parameter nor frequency space as unbounded.

## .8 Summary

This document has examined two historically and physically profound problems and shown that T0 theory resolves both through a single geometric mechanism.

**The first problem: the ultraviolet catastrophe.** Classical theory predicts infinite energy density because it tacitly assumes that frequency space is unbounded from above. Planck suppresses the divergence through quantization — but without physical justification for the upper limit. T0 eliminates the divergence fundamentally: the sub-Planck scale $L_0 = \xi \cdot \ell_P$ sets a maximum frequency $f_{\text{max}}$, above which spacetime carries no electromagnetic degrees of freedom.

**The second problem: the origin of $\epsilon = hf$.** Planck postulated the equation as an auxiliary device. Einstein forced it thermodynamically. But both left $h$ as a free parameter. T0 explains the origin: the time-field binding condition $T \cdot E = 1$ fixes directly for massless fields $E = 1/T = f \cdot \hbar/\xi$. The value of $h$ emerges from $\xi$.

**The common cause.** Both problems share the same root: classical physics treats spacetime as a continuous medium without geometric minimum structure. T0 shows that this assumption is false — and that its correction through the single parameter $\xi = 4/30000$ eliminates all anomalies.

# Bibliography

[1] A. Einstein, On a heuristic viewpoint concerning the production and transformation of light, *Annalen der Physik* **17**, 132–148 (1905).

[2] M. Planck, On the theory of the law of energy distribution in the normal spectrum, *Verhandlungen der Deutschen Physikalischen Gesellschaft* **2**, 237–245 (1900).

[3] R. A. Millikan, A direct photoelectric determination of Planck's $h$, *Physical Review* **7**, 355–388 (1916).

[4] J. Pascher, The Geometric Origin Parameter $\xi = 4/30000$, T0 Documentation Series, Document 009 (2025).

[5] J. Pascher, T0 Theory: Time-Mass Duality as a Fundamental Principle, T0 Documentation Series, Document 003 (2025).

[6] J. Pascher, Sub-Planck Structure and Geometric Regularisation in T0, T0 Documentation Series, Document 145 (2025).

[7] J. Pascher, Fractal Corrections and the g-2 Anomaly (Rev. 12), T0 Documentation Series, Document 018 (2025).

[8] J. Pascher, T0 Quantum Mechanics: Geometric Formalism and Optimisation Strategies, T0 Documentation Series, Document 034 (2025).